Python Programming (CST-005) - Unit 2

Unit 2 - Operators & Control Flow - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: operators and control flow - decisions and loops.

Key ideas: Operators: arithmetic (+ - * / // % **), comparison (== != < > <= >=), logical (and or not), assignment (+= -= *= ...), membership (in), identity (is). Watch the difference: **/** always gives float, **//** floors, ****** is power; **==** compares values, **is** compares identity. Control flow: **if** / **elif** / **else**; loops: **for** (over a sequence or range()) and **while** (condition-based). Loop control: **break** exits, **continue** skips to the next pass, and the loop-**else** runs only when the loop was not broken. range(start, stop, step) excludes the stop value.

Exam pointers: trace-the-output snippets with break/continue/else and nested loops are the usual questions; the full operator table in the notes below covers the table-format question. Practise writing: factorial, prime check, Fibonacci up to n terms, and a simple pattern print.

Extra note - Python Operators - the full table, made simple:

Type	Operators	Small example
Arithmetic	+ - * / // % **	`7 // 2 = 3`, `2 ** 3 = 8`
Comparison	== != > < >= <=	`5 > 3 = True`
Assignment	= += -= *= /=	`x += 2` means `x = x + 2`
Logical	and or not	`True and False = False`
Bitwise	& | ^ ~ << >>	`5 & 3 = 1` (bits ANDed)
Membership	in, not in	`'a' in 'anas' = True`
Identity	is, is not	`x is None`

Precedence (what runs first): brackets `()` first → power `**` → multiply/divide `* / // %` → plus/minus `+ -` → comparison → not → and → or last.

> *Just like BODMAS - brackets first, then power, then multiply-divide, then add-subtract.*

Exam tip: `//` is floor division (drops the decimal), `**` is power, and `is` checks identity while `==` checks value - these three confusions come up in every paper.

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